SOFTWARE ENGINEERING PROJECT PROPOSAL

Project Title	Sentra-X: An Evidence-Grounded and Uncertainty-Aware Human-in-the-Loop SOC Alert Triage Copilot		
Abstract	Security Operations Centers face critical alert fatigue from high-volume SIEM platforms such as Wazuh. Analysts must manually inspect raw logs to judge whether an alert is a genuine threat or a false positive—a slow, repetitive process prone to fatigue-driven errors. Existing LLM-based alert triage solutions suffer from overconfidence, hallucinated threat attributions, and vulnerability to adversarial prompt injections in log data. Sentra-X introduces an evidence-grounded, human-in-the-loop triage copilot positioned between Wazuh and the security analyst. Rather than forcing a binary classification, Sentra-X compiles structured Evidence Packs (aggregating alert telemetry, temporal log context, asset criticality, and host history) to feed a guarded LLM triage engine. The system introduces four primary contributions: (1) Explicit Evidence Grounding, requiring the LLM to cite concrete telemetry keys supporting its assessment; (2) Uncertainty-Aware Abstention, routing cases to "Needs Analyst Review" when logs are incomplete or conflicting; (3) High-Severity Safety Floor, deterministically preventing critical SIEM alerts from being downgraded by AI inference; and (4) In-Context Feedback Calibration, storing analyst overrides and dynamically retrieving them as few-shot exemplars to calibrate future triage prompts without model fine-tuning. The analyst retains complete decision authority via a web dashboard, ensuring a safe, auditable, and production-ready triage-assist system.		
Group Members	**Roll No.**	**Name**	**Signature**
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Tools to be used	• Wazuh (v4.x) - SIEM/XDR platform for generating and collecting security alerts • Python (FastAPI) - Backend service for ingestion, safety guardrail enforcement, and REST APIs • Hosted LLM API (Claude/Gemini) - Structured reasoning engine for evidence evaluation and explanation • PostgreSQL - Storage for alerts, immutable decision audits, and the few-shot feedback cache • React.js (Vite) - Web dashboard for queue inspection, evidence review, and analyst decision capture • Docker / Docker Compose - Containerized deployment of Wazuh, backend, database, and dashboard • Penpot - Wireframing and UI/UX mockups for the analyst dashboard		

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